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Principles of Teamwork

Software is a Teamwork Product

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- One of the factors that differentiates a software engineer from other conventional is engineering careers is that the duties is not limited to technical skill in design and building.
- An equally important portion of software engineers' duties relates to:
  - Analytical skill
  - Communication skill
  - Process and product management
  - People management

Software is a Teamwork Product

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- Software development relies on management or dealing with 4 Ps:
  - People: customers, developers, users, mangers, etc. In general, called stakeholder
  - Process: phases of activities to convert an idea to a product
  - Project: A planed undertaking that must be delivered at a deadline to achieve a goal
  - Product: An article or substance that is manufactured, build, or refined, normally for sale.

What is Software Team

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- Development teams work together to build a software product. Key to the success of the team is to be self-organized and cross-functional.
- Team members should make a collaborative effort when comes to communication and conducting shared responsibilities.
- Every person on the team should play a part and be held accountable in making the production happen.

Team Synergy

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- Successful start up techs look for a self-managed technical team who are heartedly dedicated to their work.
- In summary, to achieve team's synergy:
  - Each team member should believe in the fact that productivity depends on collective contribution.
    - It is a wrong thinking: "this is a trivial task, and my contribution/efforts is not important"
  - Teams have all necessary skills to progress to different phases of development:
    - Analytical skill: System thinking
    - Technical skill
    - Communication skill
    - Presentation skill
    - Marketing skill
    - Teamwork skill
- In addition to different specialized skills, as whole, accountability must be a top mission for every member.

Group Constitution

Team/Group Constitution

- The constitution deals with:
  - the rights, powers and obligations of all people involved in the group.
  - Sets out clear guidelines as to what conduct and behavior is acceptable and what is unacceptable.
- The constitution cannot extend the legal capacity, rights, powers and privileges that the group would otherwise have.

Difficulties When Working as a Team

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- The biggest problem hindering group progress is misunderstanding and miscommunication. E.g.:
  - What are the goals and objectives?
  - What is required?
  - Who is responsible?
  - What do we do in given situations?
  - When should I do act?
- Another problem is conflicting expectations.
  - What do we expect to accomplish as the project proceeds?
    - Some members' understanding of the project might not be in harmony with others.

Group Constitution Contents

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- Establish common goals and standards for the project team.
  - A brief statement of the goals of the group.
    - What do we wish to accomplish? Example (As a university student): To work industrious and diligent to accomplish the requirements of the course
    - Similar to “Statement of Purpose,” but for group not project.
- What is the format of documents? Normally companies and enterprises try to use a uniform format for technical documents (e.g., Text font size should be at least 12 points, All reports must have a cover page)
- What tools will be used?
  - Example: StarUML, Visio, etc.
- Documents, assignments, working disks:
  - Where will they be stored ?
  - Who will have access?
  - Back up procedures?
- How do we contact each other?
  - phone numbers, email addresses, etc.
- Discussions with instructor should always be done as a group.

Project Coordinator

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- Each group should assign one of the members as ”Project Coordinator”, responsible for:
  - Getting assignments posted on the D2L.
  - Breaks a deadlock (when voting).
  - Responsible for smooth flow of meetings.
  - Not a dictator!